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SSC-I NATIONAL BOOK FOUNDATION FBISE

CHEMISTRY TEST - CHAPTER 18: QUALITATIVE ANALYSIS

Name:		Roll No.		Section:

Total Time: 2:30 Hour
Time Allowed: 150 Minutes

Total Marks: 150

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. Qualitative analysis tells about the _____ of a substance and no quantity.	Volume	Nature	Mass	Number	○●○○
2. Qualitative analysis methods differ in their:	Reliability	Weight	Price	Colour	●○○○
3. Tests that confirm a substance are called _____ tests.	Physical	Quantitative	Random	Confirmatory	○○○●
4. The pH of ammonia is:	Less than 7	More than 7	Equal to 7	Zero	○●○○
5. Ammonia is a _____ gas.	Basic	Acidic	Amphoteric	Neutral	●○○○
6. Ammonia is used for making _____ fertilizer.	Superphosphate	Urea	Potash	Lime	○●○○
7. Ammonia is manufactured by the _____ process.	Solvay	Contact	Bessemer	Haber	○○○●
8. The Haber process uses hydrogen and _____ gases.	Oxygen	Chlorine	Carbon	Nitrogen	○○○●
9. To detect ammonia, we use _____ litmus paper.	Green	Yellow	Blue	Red	○○○●
10. Ammonia turns moist red litmus paper:	Blue	Red	White	Yellow	●○○○
11. Litmus paper must be moistened with _____ before testing a gas.	Alcohol	Oil	Water	Acid	○○●○
12. A basic gas such as ammonia turns red litmus:	Red	Green	Blue	White	○○●○

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Questions	A	B	C	D	A B C D
13. The gas that turns lime water milky is:	Ammonia	Oxygen	Carbon dioxide	Hydrogen	○○●○
14. Lime water is a solution of:	Calcium carbonate	Calcium chloride	Calcium oxide	Calcium hydroxide	○○○●
15. The formula of lime water solute is:	$CaCO_3$	$Ca(OH)_2$	$CaCl_2$	CaO	○●○○
16. CO_2 is prepared in the lab from marble and:	Hydrochloric acid	Water	Sodium chloride	Sodium hydroxide	●○○○
17. Marble is chemically:	Calcium hydroxide	Calcium carbonate	Calcium oxide	Calcium chloride	○●○○
18. $CaCO_3 + 2HCl \rightarrow$	$CaO + HCl$	$Ca + CO_2$	$CaCl_2 + H_2$	$CaCl_2 + CO_2 + H_2O$	○○○●
19. The milky appearance of lime water is due to:	Calcium chloride	Hydrogen	Calcium carbonate	Calcium oxide	○○●○
20. $CO_2 + Ca(OH)_2 \rightarrow$	$CaCO_3 + H_2O$	$CaO + H_2$	$CaCl_2 + H_2O$	$Ca + CO_2$	●○○○
21. The test for carbon dioxide is a _____ test.	Confirmatory	Rough	Quantitative	Physical	●○○○
22. For preparing CO_2, marble pieces are placed in a:	Pipette	Conical flask	Beaker	Burette	○●○○
23. The flask for CO_2 is corked with a cork having _____ holes.	Three	Four	Two	One	○○●○
24. In CO_2 preparation, acid is added through a:	Pipette	Beaker	Burette	Thistle funnel	○○○●
25. The gas CO_2 is passed through a delivery tube into:	Brine	Acid	Lime water	Oil	○○●○
26. Chlorine is an _____ gas.	Basic	Acidic	Inert	Neutral	○●○○
27. To detect chlorine, we use damp _____ litmus paper.	Blue	Red	Green	Yellow	●○○○
28. Damp blue litmus in chlorine turns:	Blue	Green	Red then white	Black	○○●○
29. Chlorine turns blue litmus red because it forms:	Bases in water	Acids in water	Salts	Oxides	○●○○

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Questions	A	B	C	D	A B C D
30. After turning red, litmus becomes white because chlorine is a:	Salt	Reducing agent only	Bleaching agent	Base	○○●○
31. Chlorine bleaches litmus paper to:	Black	Yellow	Green	White (colourless)	○○○●
32. Hydrogen gas is detected using a:	Lighted splint	Glowing splint	Litmus paper	Lime water	●○○○
33. Hydrogen gas burns with a:	No flame	Green flame	Pop sound	Bright flame only	○○●○
34. Hydrogen is prepared by the reaction of zinc with:	Sulphuric acid	Water	Oxygen	Sodium chloride	●○○○
35. $Zn + H_2SO_4 \rightarrow$	$Zn(OH)_2 + H_2$	$ZnSO_4 + H_2$	$ZnS + H_2O$	$ZnO + H_2$	○●○○
36. Hydrogen is collected by holding the test tube _____ over the delivery tube.	Inverted	Sideways	Upright	Horizontal	●○○○
37. Hydrogen is collected in an inverted test tube because it is:	Coloured	Light	Heavy	Sticky	○●○○
38. The apparatus for hydrogen preparation uses a _____ bottle.	Wolf	Round	Plastic	Square	●○○○
39. The wolf bottle has _____ openings.	One	Four	Two	Three	○○●○
40. The apparatus is made air tight using:	Oil	Water	Tape only	Wax	○○○●
41. $2H_2 + O_2 \rightarrow$	2OH	H_4O_2	$2H_2O$	H_2O_2	○○●○
42. The pop sound is due to the reaction of hydrogen with _____ in the presence of flame.	Oxygen	Nitrogen	Carbon	Chlorine	●○○○
43. Oxygen gas is detected using a:	Lighted splint	Lime water	Glowing splint	Litmus paper	○○●○
44. In oxygen, a glowing splint:	Turns blue	Relights into flames	Pops	Goes out	○●○○
45. Oxygen helps in:	Polymerization	Combustion	Reduction	Neutralization	○●○○
46. A source of oxygen in the lab is hydrogen:	Iodide	Sulphide	Chloride	Peroxide	○○○●

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Questions	A	B	C	D	A B C D
47. The splint test for oxygen needs a well-ventilated area and safety:	Colours	Tools	Precautions	Charts	○○●○
48. Sulphur dioxide is detected with acidified:	Sodium chloride	Litmus	Potassium permanganate	Lime water	○○●○
49. SO_2 is a good _____ agent.	Oxidizing	Bleaching only	Reducing	Neutral	○○●○
50. $KMnO_4$ solution is bright:	Blue	Yellow	Green	Purple	○○○●
51. SO_2 discharges the _____ colour of $KMnO_4$.	Blue	Yellow	Purple	Green	○○●○
52. The presence of SO_2 gas turns acidified $KMnO_4$ from purple to:	Black	Colourless	Blue	Red	○●○○
53. The SO_2 test is used in analytical chemistry and _____ monitoring.	Political	Medical only	Environmental	Social	○○●○
54. In the $KMnO_4$ + SO_2 reaction, the manganese-containing product is:	$MnCl_2$	$KMnO_4$	MnO_2 (as per the book)	Mn	○○●○
55. Flame tests detect metal:	Radicals	Anions	Cations	Atoms of gases	○○●○
56. A Bunsen burner flame has two parts: oxidizing and:	Neutral	Cold	Hot	Reducing	○○○●
57. The upper part of a Bunsen flame is the _____ flame.	Oxidizing	Neutral	Luminous only	Reducing	●○○○
58. The lower part of a Bunsen flame is the _____ flame.	Oxidizing	Hot only	Reducing	Neutral	○○●○
59. Bunsen flame has enough energy to excite electrons of alkali and _____ metals.	Transition	Halogen	Noble gas	Alkaline earth	○○○●
60. Alkali metals have _____ ionization potentials.	Low	Zero	Infinite	High	●○○○
61. When excited electrons return to the ground state, they emit:	Heat only	Light	Sound	Electrons	○●○○
62. The colour of the flame is characteristic of the:	Solvent	Anion	Metal ion	Container	○○●○
63. In a flame test, the sample is held on a clean wire loop of:	Iron	Zinc	Copper	Platinum	○○○●

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Questions	A	B	C	D	A B C D
64. In a flame test, a clean _____ may also be used to hold the sample.	Plastic rod	Glass rod	Wooden rod	Iron rod	○●○○
65. The sample for a flame test is usually dissolved in:	Alcohol only	Acid only	Oil	Water	○○○●
66. The flame colour of sodium is:	Green	Golden yellow	Crimson red	Purple	○●○○
67. The flame colour of lithium is:	Green	Purple	Golden yellow	Crimson red	○○○●
68. The flame colour of strontium is:	Green	Crimson red	Golden yellow	Purple	○●○○
69. The flame colour of potassium is:	Crimson red	Purple (lilac)	Green	Golden yellow	○●○○
70. The flame colour of caesium is:	Green	Golden yellow	Blue	Crimson red	○○●○
71. The flame colour of barium is:	Crimson red	Golden yellow	Blue	Green	○○○●
72. The flame colour of copper is:	Golden yellow	Crimson red	Bluish green	Purple	○○●○
73. Which metal ion gives a golden yellow flame?	Ba^{2+}	Li^{+}	Na^{+}	K^{+}	○○●○
74. Which metal ion gives a purple flame?	Ba^{2+}	K^{+}	Na^{+}	Li^{+}	○●○○
75. Which metal ion gives a green flame?	K^{+}	Ba^{2+}	Li^{+}	Na^{+}	○●○○
76. Which metal ion gives a bluish green flame?	K^{+}	Na^{+}	Li^{+}	Cu^{2+}	○○○●
77. Which metal ion gives a crimson red flame along with lithium?	Sr^{2+}	Na^{+}	K^{+}	Ba^{2+}	●○○○
78. Both lithium and strontium give a _____ flame.	Purple	Crimson red	Green	Golden yellow	○●○○
79. Common salt (NaCl) in a flame gives a _____ colour.	Green	Purple	Red	Golden yellow	○○○●
80. Potassium chloride (KCl) in a flame gives _____ colour.	Purple	Red	Golden yellow	Green	●○○○

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Questions	A	B	C	D	A B C D
81. Barium chloride in a flame gives _____ colour.	Purple	Golden yellow	Green	Red	○○●○
82. Copper sulphate in a flame gives _____ colour.	Golden yellow	Purple	Red	Bluish green	○○○●
83. Fireworks produce colours due to metal ions similar to:	Filtration	Distillation	Titration	Flame tests	○○○●
84. Red litmus turns blue in:	Neutral gas	Acidic gas	Basic gas	Inert gas	○○●○
85. Blue litmus turns red in:	Neutral gas	Inert gas	Basic gas	Acidic gas	○○○●
86. Which gas turns lime water milky?	O_2	NH_3	CO_2	H_2	○○●○
87. Which gas gives a pop sound with a lighted splint?	NH_3	H_2	CO_2	O_2	○●○○
88. Which gas relights a glowing splint?	O_2	CO_2	H_2	NH_3	●○○○
89. Which gas turns damp blue litmus red then white?	H_2	Cl_2	CO_2	NH_3	○●○○
90. Which gas turns moist red litmus blue?	Cl_2	NH_3	CO_2	SO_2	○●○○
91. Which gas decolourizes acidified $KMnO_4$?	H_2	SO_2	NH_3	CO_2	○●○○
92. The gas test for hydrogen is called the _____ test.	Blue	Milky	Bleach	Pop	○○○●
93. The gas test for carbon dioxide is called the _____ test.	Lime water	Glowing splint	Litmus	Pop	●○○○
94. The gas test for oxygen is called the _____ test.	Pop	Litmus	Glowing splint	Lime water	○○●○
95. Which is the confirmatory test for ammonia?	Relights splint	Turns red litmus blue	Pop sound	Turns lime water milky	○●○○
96. Which is the confirmatory test for CO_2?	Pop sound	Lime water turns milky	Bleaches litmus	Relights splint	○●○○
97. Which is a confirmatory test for chlorine?	Bleaches damp litmus	Pop sound	Turns lime water milky	Relights splint	●○○○

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Questions	A	B	C	D	A B C D
98. Which is a confirmatory test for oxygen?	Relights a glowing splint	Bleaches litmus	Turns lime water milky	Pop sound	●○○○
99. Which is a confirmatory test for hydrogen?	Relights splint	Bleaches litmus	Pop sound with lighted splint	Turns lime water milky	○○●○
100. Which is a confirmatory test for SO_2?	Discharges purple $KMnO_4$	Bleaches litmus	Relights splint	Pop sound	●○○○
101. Chlorine gas is a bleaching agent because it forms _____ in water.	Sodium chloride	Hydrochloric acid only	Hypochlorous acid	Chloric acid	○○●○
102. Hydrogen is a _____ gas.	Light	Coloured	Smelly	Heavy	●○○○
103. Marble chips are placed in the flask for the _____ preparation.	O_2	NH_3	CO_2	H_2	○○●○
104. Zinc pieces are placed in the wolf bottle for _____ preparation.	H_2	O_2	NH_3	CO_2	●○○○
105. Lime water turns milky due to the formation of insoluble:	$Ca(OH)_2$	CaO	$CaCO_3$	$CaCl_2$	○○●○
106. The unit before Qualitative Analysis in the book is:	Separation Techniques	Empirical Data Collection and Analysis	Biochemistry	Hydrocarbons	●○○○
107. The unit after Qualitative Analysis in the book is:	Separation Techniques	Hydrocarbons	Biochemistry	Chromatography	○○○●
108. Qualitative analysis answers the question:	How fast?	How heavy?	What is present?	How much is present?	○○●○
109. Quantitative analysis answers the question:	How much is present?	What is present?	What colour?	What smell?	●○○○
110. The test for CO_2 requires the gas to be passed through:	Brine	Oil	Acid	Lime water	○○○●
111. The test for ammonia is performed at the mouth of a test tube containing:	CO_2	Ammonia gas	O_2	H_2	○●○○
112. A gas that supports burning is:	Hydrogen	Carbon dioxide	Ammonia	Oxygen	○○○●
113. A gas that burns with a pop sound is:	Oxygen	Ammonia	Carbon dioxide	Hydrogen	○○○●
114. A gas that extinguishes a lighted splint and turns lime water milky is:	Hydrogen	Ammonia	Oxygen	Carbon dioxide	○○○●

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Questions	A	B	C	D	A B C D
115. A colourless gas with a pungent smell that turns red litmus blue is:	Ammonia	Hydrogen	CO_2	Chlorine	●○○○
116. A greenish-yellow gas that bleaches damp litmus is:	Chlorine	Ammonia	CO_2	Hydrogen	●○○○
117. The flame test is done by heating the sample on a wire in the:	Water	Bunsen flame	Oil	Ice	○●○○
118. The flame test works because metal ions emit _____ when electrons fall back.	Protons	Visible light	Sound	Electrons	○●○○
119. Atoms get excited by absorbing:	Mass	Charge	Energy	Colour	○○●○
120. Sodium's flame colour is also seen in:	Street lamps	Stars only	Candles	Rainbows	●○○○
121. The wolf bottle apparatus is made air tight so that the gas does not:	Freeze	Dissolve	Burn	Leak	○○○●
122. A reference chart is used in flame tests to _____ the colour of the flame.	Hide	Dissolve	Compare	Burn	○○●○
123. A test that tells the identity of a substance but not its amount is:	Qualitative	Quantitative	Chemical only	Physical	●○○○
124. Which litmus is used to test a basic gas like ammonia?	Blue litmus	Red litmus	Green litmus	No litmus	○●○○
125. Which litmus is used to test an acidic gas like chlorine?	Blue litmus	No litmus	Green litmus	Red litmus	●○○○
126. A glowing splint is a wooden splint with a glowing:	Handle	Middle	Base	Tip	○○○●
127. A lighted splint is a wooden splint that is:	Cold	Wet	Glowing only	Burning with a flame	○○○●
128. Which splint is used for the hydrogen test?	Lighted splint	Glowing splint	Wet splint	Cold splint	●○○○
129. Which splint is used for the oxygen test?	Glowing splint	Wet splint	Lighted splint	Cold splint	●○○○
130. Hydrogen peroxide when poured into a test tube generates:	Oxygen	Chlorine	Ammonia	Hydrogen	●○○○
131. Dilute sulphuric acid with zinc gives hydrogen and zinc:	Sulphate	Nitrate	Oxide	Chloride	●○○○

Questions	A	B	C	D	A B C D
132. Gas jars for gas tests should be handled in a _____ area.	Dark	Hot	Closed	Well-ventilated	○○○●
133. The pungent gas with the smell of cleaning agents is:	CO_2	Ammonia	Hydrogen	Oxygen	○●○○
134. Which gas is a pollutant detected by $KMnO_4$?	Oxygen	Sulphur dioxide	Nitrogen	Hydrogen	○●○○
135. The purple colour of $KMnO_4$ is due to the ion:	MnO_4^-	Mn^{2+}	Mn^{3+}	MnO_2	●○○○
136. In flame tests, the colour comes from electrons falling to the:	Ground state	Excited state	Nucleus	Next shell up	●○○○
137. In flame tests, electrons are first excited to a _____ energy level.	Negative	Higher	Zero	Lower	○●○○
138. Caesium gives a _____ flame.	Yellow	Red	Green	Blue	○○○●
139. Which of these gives crimson red flame?	Barium	Sodium	Potassium	Lithium	○○○●
140. Which of these gives a green flame?	Potassium	Lithium	Sodium	Barium	○○○●
141. A flame test is mainly used for which metals?	Noble gases	Halogens	Non-metals	Alkali and alkaline earth metals	○○○●
142. Metal ions are identified in a flame test by comparing the colour with a:	Clock	Reference chart	Ruler	Balance	○●○○
143. The sample for a flame test can be a salt, oxide or any compound containing the:	Halogen	Non-metal	Metal ion	Noble gas	○○●○
144. The wire loop in a flame test must be clean and:	Reactive	Non-reactive	Wet	Rusty	○●○○
145. Using an unclean wire loop in a flame test would give:	A bigger flame	No flame	Right colour	Wrong colour	○○○●
146. The Bunsen burner flame has enough energy to excite alkali metals because they have:	Low ionization energy	Many protons	High ionization energy	No electrons	●○○○
147. The characteristic flame colour is a property of the _____ ion.	Metal	Liquid	Gas	Non-metal	●○○○
148. When a gas test uses lime water, the reagent must be:	Milky	Frozen	Clear	Hot	○○●○

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Questions	A	B	C	D	A B C D
149. Passing excess CO_2 through lime water eventually makes it:	Colder	Hotter	Blacker	Clear again	○○○●
150. The conical flask used in gas preparation is sealed with a:	Tape	Cork	Rubber band	Plastic wrap	○●○○

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